

Generation Interconnection System Impact Study Report

For Queue Project Q15-0424

Glen Park Hydroelectric Project 69 kV

January 28, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0424. The Interconnection Customer proposes a 890.1 MW natural gas combined cycle facility interconnecting at the Glen Park Hydroelectric Project 69 kV substation in the North Carolina area. The request was received on January 17, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 177 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$129,993,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N PASCO 138/69 kV Auto	M.L. HIBBARD S.E. STA 138/1 kV TX	647.4	52	40.9
M.L. HIBBARD S.E. STATION 138 kV/1 kV Auto	N PASCO 138 kV/69 kV TX	188.7	229	52.5
TRAVIS AB – WERNER WEST 138kV	ROANOKE RAPIDS – EDGEcombe GENCO LLC 345kV	165.2	259	34.1
SOUTHAMPTON POWER STATION – N PASCO 138kV Line	M.L. HIBBARD S.E. STA 138 kV/1 kV TX	158.7	203	40.9
SOUTHAMPTON POWER STA – TRAVIS AB 138kV Line	M.L. HIBBARD S.E. STA 138/1 kV Transformer	145.4	212	40.9
FLORADAY – VILLA DEL REY 3010 138kV	MCCARTNEY – N TOWER TAP 345kV Line	125.2	289	10.6
N TOWER TAP 345/138kV Auto	NORTH TOWER TAP – HAGGIN 345kV Line	121.8	467	13.5
DEEN PT – LOCKS 138kV	MCCARTNEY – N TOWER TAP 345kV	121.1	169	7.8
GASTON – E SPARTA 138kV	ROANOKE RAPIDS – EDGEcombe GENCO LLC 345kV	120.2	204	12.4
WALTHALL – UNKNOWN SUBSTATION 138kV	NORTH TOWER TAP – HAGGIN 345kV L	118.7	317	9.6
PLANO PARKER RD 138/69kV Transformer Bank	EDGEcombe GENCO LLC – ROANOKE RAPIDS 345kV	117.7	87	5.4
FLORADAY – SONOMA 3 138kV	N TOWER TAP – MCCARTNEY 345kV	117.5	203	7.8
SONOMA 3 SUBSTATION – LOCKS 138kV L	MCCARTNEY – N TOWER TAP 345kV	117.1	203	7.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NORTH TOWER TAP – UNKNOWN SUBSTATION 138kV	NORTH TOWER TAP – HAGGIN 345kV	114.0	344	9.6
N TOWER TAP – DPC ELROY 138kV	MCCARTNEY – N TOWER TAP 345kV Line	112.0	286	6.1
HULL ST RD SUBSTATION 345-138 kV XFMR	MITCELL DAM – SKAGGS ISLAND JCT 2 345kV Line	110.6	529	13.3
JIM HILL – HAGGIN 138kV	MITCHELL DAM – SKAGGS ISLAND JUNCTION 2 345kV	109.7	257	5.7
YORK CANYON – HAT RAPIDS 138kV Line	MEAUX – WHITE STONE 138kV	109.0	589	6.5
DALEVILLE – LAMPING STA 138kV	NEW DENVER – GREENING 138kV	108.6	556	6.2
POTOSI SUBSTATION – TYLER SUBSTATION 138kV	MITCHELL DAM – SKAGGS ISLAND JUCTION 2 345kV	107.3	269	5.6
GASTON – RACINE 138kV	N TOWER TAP – MCCARTNEY 345kV	106.5	246	6.5
DELTAPORT – POTOSI 138kV Line	MITCHELL DAM – SKAGGS ISLAND JUNCTION 2 345kV	106.4	246	5.6
FYFFE 138/1 kV XFMR	M.L. HIBBARD S.E. STA SUBSTATION 138/1kV TX	105.9	238	34.2
EMPORIA – FLAT TOP 138kV	MCCARTNEY – N TOWER TAP 345kV line	104.5	232	5.0
LAWRENCEVILLE – RACINE SUBSTATION 138kV	MCCARTEY – N TOWER TAP 345kV line	104.1	239	6.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NORTH PASCO 138 kV/69 kV Transformer	Replace transformer with higher-rated unit	\$3,801,000	95% (\$3,620,000)
M.L. HIBBARD S.E. STATION 138 kV/1 kV Auto	Replace transformer with higher-rated unit	\$4,929,000	100% (\$4,929,000)
TRAVIS AB – WERNER WEST 138kV	Rebuild line with higher-capacity conductor	\$8,113,000	94% (\$7,599,000)
SOUTHAMPTON POWER STATION – N PASCO 138kV Line	Rebuild line with higher-capacity conductor	\$30,613,000	95% (\$29,153,000)
TRAVIS AB – SOUTHAMPTON POWER STA 138kV Line	Rebuild line with higher-capacity conductor	\$5,432,000	95% (\$5,173,000)
FLORADAY – VILLA DEL REY 3010 138kV	Rebuild line with higher-capacity conductor	\$5,952,000	32% (\$1,887,000)
N TOWER TAP 345/138kV Auto	Replace transformer with higher-rated unit	\$14,429,000	30% (\$4,335,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LOCKS SUBSTATION – DEEN PT 138kV	Rebuild line with higher-capacity conductor	\$1,766,000	30% (\$527,000)
GASTON – E SPARTA 138kV	Rebuild line with higher-capacity conductor	\$33,480,000	100% (\$33,480,000)
WALTHALL – UNKNOWN SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$12,984,000	32% (\$4,166,000)
PLANO PARKER RD 138 kV/69 kV Autotransformer	Replace transformer with higher-rated unit	\$4,911,000	100% (\$4,911,000)
FLORADAY – SONOMA 3 138kV	Rebuild line with higher-capacity conductor	\$16,089,000	30% (\$4,801,000)
SONOMA 3 SUBSTATION – LOCKS 138kV L	Rebuild line with higher-capacity conductor	\$2,388,000	30% (\$713,000)
NORTH TOWER TAP – UNKNOWN 138kV Line	Rebuild line with higher-capacity conductor	\$4,803,000	32% (\$1,541,000)
N TOWER TAP – DPC ELROY 138kV	Rebuild line with higher-capacity conductor	\$6,414,000	100% (\$6,414,000)
HULL STREET ROAD 345/138kV Auto	Replace transformer with higher-rated unit	\$16,319,000	22% (\$3,567,000)
JIM HILL – HAGGIN 138kV Line	Reconductor line	\$332,000	25% (\$84,000)
YORK CANYON – HAT RAPIDS SUBSTATION 138kV L	Reconductor line	\$13,495,000	7% (\$983,000)
DALEVILLE – LAMPING STA 138kV	Reconductor line	\$1,072,000	9% (\$93,000)
POTOSI SUBSTATION – TYLER SUBSTATION 138kV	Reconductor line	\$173,000	25% (\$43,000)
GASTON – RACINE 138kV	Reconductor line	\$4,062,000	70% (\$2,842,000)
DELTAPORT – POTOSI 138kV	Reconductor line	\$1,316,000	25% (\$330,000)
FYFFE 138/1 kV XFMR	Replace transformer with higher-rated unit	\$6,108,000	95% (\$5,822,000)
EMPORIA – FLATTOP 138kV	Reconductor line	\$876,000	82% (\$722,000)
RACINE – LAWRENCEVILLE 138kV L	Reconductor line	\$3,228,000	70% (\$2,258,000)
	Total allocated to project		\$129,993,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.